Polymorphism

22 March 2024 23:12

1. Method Overloading: ← has multiple methods having same name but different in parameters, it is known as Method Overloading.

Different ways to overload a method:

1. By changing the number of arguments.
2. By changing the data type.

```
package Polymorphism;

/**
 * Write a description of class Shapes here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Shapes
{
   /**
    * returns the area of circle in double.
    * @param radius of a circle
    */
   public double area(double radius){
      return Math.PI * radius * radius;
   }

   /**
    * returns the area of rectangle in double
    * @param length length of the rectangle
    * @param breadth breadth of the rectangle
    */
   public double area(double length, double breadth){
      return length * breadth;
   }
}
```

1 double args.

2 double args.

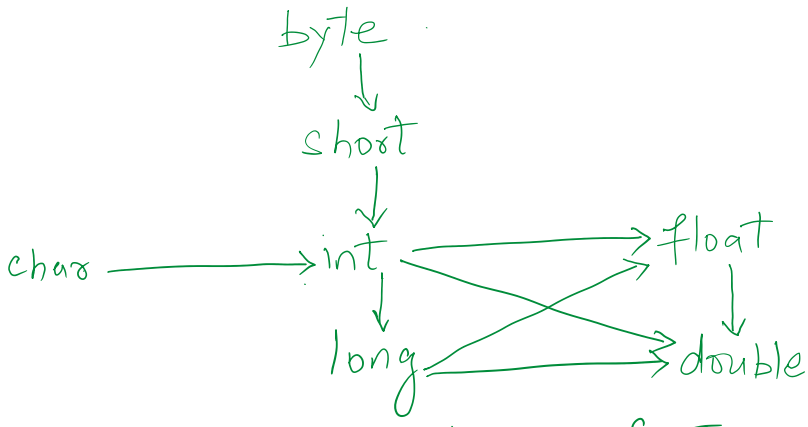
Point-2

```
package Polymorphism;

/**
 * Write a description of class Shapes here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Shapes
{
   /**
    * returns the area of circle in double.
    * @param radius of a circle
    */
   public double area(double radius){
      return Math.PI * radius * radius;
   }

   /**
    * returns the area of rectangle in double
```

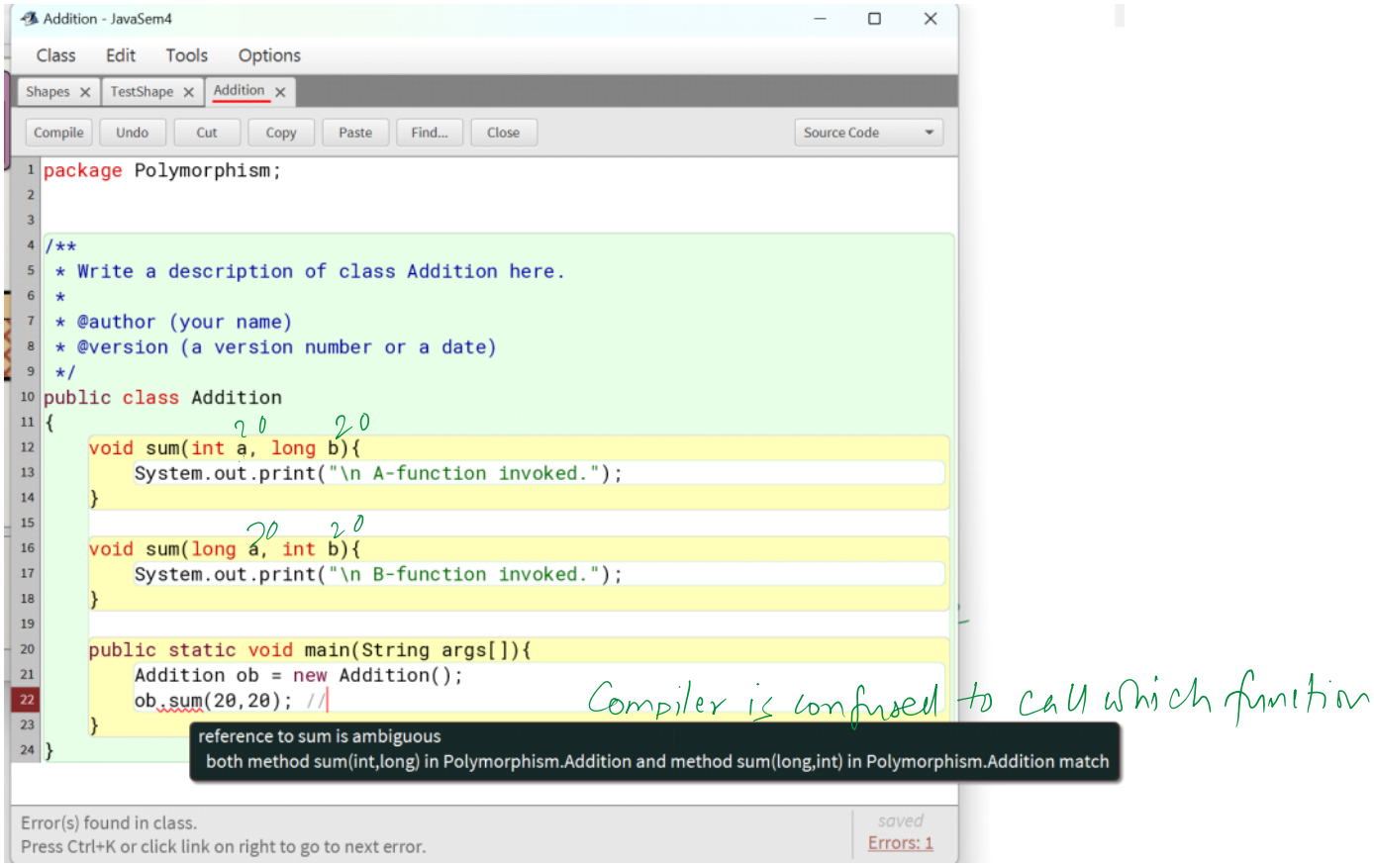
```
     * @param length length of the rectangle
     * @param breadth breadth of the rectangle
     */
    public double area(double length, double breadth){
        return length * breadth;
    }
    /**
     * Overloaded area method which returns area in int form.
     * @param length length of the rectangle
     * @param breadth breadth of the rectangle
     */
    public int area(int len, int breadth){
        return len * breadth;
    }
}
```

different data types in the parameter.

Method Overloading and Type Promotion.



byte can be promoted to short, int, long, float or double
short can " " " int, long float or double.
int " " " " long, float or double.
long " " " " float, double.
float " " " " double only.




```
package Polymorphism;


/**
 * Write a description of class Addition here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Addition
{
   void sum(int a, long b){
      System.out.print("\n A-function invoked.");
   }

   void sum(long a, int b){
      System.out.print("\n B-function invoked.");
   }

   public static void main(String args[]){
      Addition ob = new Addition();
      //ob.sum(20,20); //case of ambiguity
      ob.sum(20,20L); // to remove ambiguity we need to specify proper data-types.
      ob.sum(30L,30);
   }
}
```